

lindblad equation steady state

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1 Lindblad equation

One can find the derivation of the Lindblad equation in [3] Given $\rho \in B(\mathcal{H})$ be density matrix of dimension n . The Lindblad master equation is given as follows,

$$\dot{\rho}(t) = -\frac{i}{\hbar}[H, \rho] + \sum_{i=1}^{n^2-1} \gamma_i (V_i \rho V_i^* - \frac{1}{2}\{V_i^* V_i, \rho\})$$

where $H = H^*$ and $V_i \in B(\mathcal{H})$

Now, for simplicity, we will consider the case when $n = 2$. In this case, we know one of the basis of $B(\mathcal{H})_{sa}$ is Pauli matrices, i.e.

$$\begin{aligned}\sigma_0 &= I_2 \\ \sigma_x &= \sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ \sigma_y &= \sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \\ \sigma_z &= \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\end{aligned}$$

We first consider the representation of Lindblad master equation using Pauli matrices. Note that any Hermitian matrix is of the form $a_0\sigma_0 + a_1\sigma_1 + a_2\sigma_2 + a_3\sigma_3$ where $a = (a_0, a_1, a_2, a_3) \in \mathbb{R}^4$. Also, we denote $\sigma = (\sigma_1, \sigma_2, \sigma_3)$.

Then, we assume $H = h'I + h \cdot \sigma$ and $\rho = r'I + r \cdot \sigma$. Consider the commutator term,

$$\begin{aligned}[H, \rho] &= [h'I + h \cdot \sigma, r'I + r \cdot \sigma] = [h'I, r'I] + [h'I, r \cdot \sigma] + [h \cdot \sigma, r'I] + [h \cdot \sigma, r \cdot \sigma] \\ &= [h \cdot \sigma, r \cdot \sigma] = 2i(h \times r) \cdot \sigma\end{aligned}$$

The last term is obtained through direct computation, and the cross notation represents the outer product.

Next, we will compute the Lindblad operator term. The derivation is as follows,

$$V_i \rho V_i^* - \frac{1}{2} \{V_i^* V_i, \rho\} = V_i(r' I + r \cdot \sigma) V_i - \frac{1}{2} \{V_i^* V_i, r' I + r \cdot \sigma\} = r' [V_i, V_i^*] + V_i(r \cdot \sigma) V_i^* - \frac{1}{2} \{V_i^* V_i, r \cdot \sigma\}$$

Now, we will consider the decomposition $V_i = A + iB$ where $A = a' I + a \cdot \sigma$, $B = b' I + b \cdot \sigma$ are both hermitian, and the latter term will becomes,

$$\begin{aligned} V_i(r \cdot \sigma) V_i^* - \frac{1}{2} \{V_i^* V_i, r \cdot \sigma\} &= \frac{1}{2} ([A, [r \cdot \sigma, A]] + [B, [r \cdot \sigma, B]] - i[A, \{r \cdot \sigma, B\}] + i[B, \{r \cdot \sigma, A\}]) \\ &= \frac{1}{2} (-4a \times (r \times a) \cdot \sigma - 4b \times (r \times b) \cdot \sigma + 4b'(a \times r) \cdot \sigma - 4a'(b \times r) \cdot \sigma) \\ &= -2(a \times (r \times a) + b \times (r \times b) - b'(a \times r) + a'(b \times r)) \cdot \sigma \end{aligned}$$

Then, combining with

$$r' [V_i, V_i^*] = r' ((A + iB)(A - iB) - (A - iB)(A + iB)) = -2ir' [A, B] = -2ir' [a \cdot \sigma, b \cdot \sigma] = 4r'(a \times b) \cdot \sigma$$

We have the following identity,

$$V_i \rho V_i^* - \frac{1}{2} \{V_i^* V_i, \rho\} = -2(a \times (r \times a) + b \times (r \times b) - b'(a \times r) + a'(b \times r)) \cdot \sigma + 4r'(a \times b) \cdot \sigma$$

Thus, introducing the index, the Lindblad equation can be reconstructed as,

$$\begin{aligned} \dot{\rho} &= \frac{2}{\hbar} (h \times r) \cdot \sigma + \sum_{i=1}^3 -2\gamma_i (a_i \times (r \times a_i) + b_i \times (r \times b_i) - b'_i (a_i \times r) + a'_i (b_i \times r)) \cdot \sigma + 4r' \gamma_i (a_i \times b_i) \cdot \sigma \\ &= \frac{2}{\hbar} O(h) r \cdot \sigma - 2 \sum_{i=1}^3 \gamma_i M(V_i) r \cdot \sigma + \sum_{i=1}^3 \gamma_i 4r' (a_i \times b_i) \cdot \sigma \end{aligned}$$

where $M(V_i) = |a_i|^2 + |b_i|^2 - a_i a_i^* - b_i b_i^* + a'_i O(b_i) - b'_i O(a_i)$. Note that since cross product is a linear operation, we can define $O(b)r := b \times r$. Also, we have used the property of $a \times (b \times c) = b(a \cdot c) - c(a \cdot b)$ to derive $M(V_i)$.

Since ρ is density matrix, $r' = \frac{1}{2}$ and $\dot{\rho}(t) = \dot{r}(t) \cdot \sigma$ and because $\{I, \sigma_1, \sigma_2, \sigma_3\}$ are the basis, we have the following ode equation,

$$\dot{r} = \frac{2}{\hbar} O(h) r - 2 \sum_{i=1}^3 \gamma_i M(V_i) r + \sum_{i=1}^3 2\gamma_i (a_i \times b_i) \quad (1)$$

To better see the structure, lets define $\mathbf{D} := 2(\frac{1}{\hbar} O(h) - \sum_{i=1}^3 \gamma_i M(V_i))$ and $C = \sum_{i=1}^3 \gamma_i (a_i \times b_i)$. Now, rewrite the equation, we get

$$\dot{r} = \mathbf{D} r + 2C$$

2 Steady states

Proposition 2.1. *The qubit Lindblad master equation has a unique steady state density matrix ρ_∞ if and only if the Bloch form equation, $\dot{r} = \mathbf{D}r + 2C$ has an invertible matrix $\mathbf{D}$.*

Proof. □

Next, we will find the kernel of $\mathbf{D}$.

Lemma 2.1. *Let S be a skew-symmetric matrix and H be a Hermitian negative semidefinite matrix. Then*

$$\ker(S + H) = \ker S \cap \ker H.$$

Proof. Let $x \in \ker(S + H)$. Then

$$(S + H)x = 0 \quad \Rightarrow \quad Sx = -Hx.$$

Taking the inner product with x , we obtain

$$0 = \langle (S + H)x, x \rangle = \langle Sx, x \rangle + \langle Hx, x \rangle.$$

Since S is skew-symmetric, $\langle Sx, x \rangle \in i\mathbb{R}$, and since H is Hermitian negative semidefinite, $\langle Hx, x \rangle \in \mathbb{R}_{\leq 0}$. Hence both terms must vanish, and in particular

$$\langle Hx, x \rangle = 0.$$

Because $H \leq 0$, this implies $Hx = 0$, and consequently $Sx = 0$ from $Sx = -Hx$. Thus $x \in \ker S \cap \ker H$.

The converse inclusion is immediate, so the claim follows. □

We decompose the operator $\mathbf{D}$ as

$$\mathbf{D} = A + S,$$

where

$$S = 2 \sum_{i=1}^3 \gamma_i (a_i a_i^* + b_i b_i^* - |a_i|^2 I - |b_i|^2 I), \quad A = 2O \left(\frac{1}{h} h + \sum_{i=1}^3 \gamma_i (b_i' a_i - a_i' b_i) \right).$$

From construction, A is skew-symmetric and S is Hermitian negative semidefinite (the latter follows from the Cauchy-inequality). Therefore, by the lemma,

$$\ker \mathbf{D} = \ker A \cap \ker S.$$

For the skew operator $O(v)$, we have $\ker O(v) = \text{span}\{v\}$, and hence

$$\ker A = \text{span} \left\{ \frac{1}{h} h + \sum_{i=1}^3 \gamma_i (b_i' a_i - a_i' b_i) \right\}.$$

Moreover, for any nonzero vector a ,

$$\ker(aa^* - |a|^2 I) = \text{span}\{a\}.$$

Consequently,

$$\ker S = \bigcap_{i=1}^3 \text{span}\{a_i\} \cap \bigcap_{i=1}^3 \text{span}\{b_i\}.$$

Thus, $\ker \mathbf{D} = \{0\}$ if there does not exist nonzero coefficients $\{c_i\}_{i=1}^7$ such that $c_1 a_1 = c_2 b_1 = \dots = c_5 a_3 = c_6 b_3 = c_7 h$ (There is another situation that $\frac{1}{h}h + \sum_{i=1}^3 \gamma_i(b'_i a_i - a'_i b_i) = 0$ can lead to the conclusion), or in operator sense, there does not exist any nonzero vector $v \in \mathbb{R}^3$ such that $[\Re V_1, v \cdot \sigma] = [\Im V_1, v \cdot \sigma] = \dots = [H, v \cdot \sigma] = 0$. Thus, the condition is equivalent to there does not exist any nonzero vector $[V_1, v \cdot \sigma] = [V_1^*, v \cdot \sigma] = \dots [H, v \cdot \sigma] = 0$, meaning that $\{H, V_1, V_2, V_3, V_1^*, V_2^*, V_3^*\}' = \mathbb{C}I$ (This is because in qubit case, any hermitian can be represented using pauli basis), which is equivalent to Frigerio's theorem[2] in qubit case.

So far we have found the condition for the uniqueness of the steady state. We will next compute the convergence rate given that the steady state, as we denoted by $r_\infty = -2\mathbf{D}^{-1}C$, is unique.

To see the convergence rate, we compute the bound of $\|r - r_\infty\|$. Notice that $\frac{d}{dt}(r - r_\infty) = \dot{r} = \mathbf{D}(r - r_\infty)$. Integrating from both side, we have $r - r_\infty = e^{\mathbf{D}t}(r(0) - r_\infty)$. Consequently,

$$\|r - r_\infty\| = \|e^{\mathbf{D}t}(r(0) - r_\infty)\| \leq \|e^{\mathbf{D}t}\| \|r(0) - r_\infty\|$$

For now, we will evaluate the spectrum of $\mathbf{D} = A + S$. From the previous part, we have known that $\mathbf{D}$ is composed of skew symmetric and negative semi-definite part. Then,

$$\Re \langle \mathbf{D}x, x \rangle = \frac{1}{2}(\langle \mathbf{D}x, x \rangle + \langle \mathbf{D}^*x, x \rangle) = \frac{1}{2}(\langle (A + S + A^* + S^*)x, x \rangle) = \langle Sx, x \rangle$$

That is, for any $\lambda \in \sigma(\mathbf{D})$,

$$\Re \lambda \leq \sup_{\|x\|=1} \langle Sx, x \rangle \leq 0$$

Thus, the convergence rate $\|r - r_\infty\| \leq Ce^{-\mu t}$, where $\mu = -\sup_{\|x\|=1} \langle Sx, x \rangle = \inf_{\|x\|=1} \langle -Sx, x \rangle [1]$. Much Specifically,

$$\mu = 2 \inf_{\|x\|=1} \sum_{i=1}^3 \gamma_i \langle K_i x, x \rangle = 2 \inf_{\|x\|=1} \sum_{i=1}^3 \gamma_i (\|a_i \times x\|^2 + \|b_i \times x\|^2)$$

where $K_i = |a_i|^2(I - P_{a_i}) + |b_i|^2(I - P_{b_i})$ (The last equality only serves as some geometric intuition, because x may be in $\mathbb{C}^3$). What is the lower bound for μ ?

A simple yet worthy of discussion example is that let a_i, b_j parallel to each other, and let h out of parallel. Then, it satisfy Frigerio's condition. However, $\mu = \inf_{\|x\|=1} C \langle (I - P_{a_1})x, x \rangle = 0$. This case clearly illustrates the unsatisfactory results for Frigerio's condition, the best we can say so far is $\mu \geq 0$. We need more condition to find the good convergent rate.

References

- [1] Exponential bound on norm of matrix exponential (of linear ode). *Mathematics Stack Exchange*, 2017.
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- [3] Daniel Manzano. A short introduction to the lindblad master equation. 2020.